

a) $L(x, y)$: person x has lived in place y .

Domain: Students of the school

A student in your place has lived in Vietnam.

$\exists x L(x, \text{Vietnam})$

Domain: All people. $\exists x (S(x) \wedge L(x, \text{Vietnam}))$

$V(x)$: Person x has lived in Vietnam

Domain: Students of the school

$\exists x V(x)$

Domain: All people $\exists x (S(x) \wedge V(x))$

b) There is a student in your school who cannot speak Hindi

$H(x)$: Person x can speak Hindi

Domain: Students of the school $\exists x (\neg H(x))$

Domain: All people

$\exists x (S(x) \wedge \neg H(x))$

$L(x, y)$: Person x can speak language y .

Domain: Students of the school $\exists x (\neg L(x, \text{Hindi}))$

Domain: All people

~~$\exists x (S(x) \wedge \neg L(x, \text{Hindi}))$~~ $\exists x (S(x) \wedge \neg L(x, \text{Hindi}))$

c) A student in your school knows Java, Prolog and C++.

$J(x)$: Person x knows Java

$P(x)$: Person x knows Prolog

$C++(x)$: Person x knows C++.

Domain: Students of the school

Domain: All people

$\exists x (J(x) \wedge P(x) \wedge C++(x))$

$\exists x (S(x) \wedge J(x) \wedge P(x) \wedge C++(x))$

$PL(x, y)$: Person x knows programming language y .

Domain: Students of the school $\exists x (PL(x, \text{Java}) \wedge PL(x, \text{Prolog}) \wedge PL(x, \text{C++}))$

Domain: All people $\exists x (S(x) \wedge PL(x, \text{Java}) \wedge PL(x, \text{Prolog}) \wedge PL(x, \text{C++}))$